

Relations and their properties

(Paragraph 9.1)

Let A and B be sets. A binary relation from A to B is a subset R of $A \times B$. We use the notation $a R b$ to denote that $(a, b) \in R$ and $a \not R b$ to denote that $(a, b) \notin R$.

Moreover, when (a, b) belongs to R , a is said to be related to b by R .

A is the set of students in the school.

B is the set of courses.

Let's R consists of all pairs (a, b) , where student a is enrolled in the course b .

Assume that Goodfriend and Sherman are enrolled in the course CS518.

Then we write $(\text{Goodfriend}, \text{CS518}) \in R$ and $(\text{Sherman}, \text{CS518}) \in R$

or Goodfriend R CS518 and Sherman R CS518.

$(\text{Sherman}, \text{CS510}) \notin R$ means that Sherman does not enrolled in the course CS510.

Similarly for Sherman $\not R$ CS510.

Let $R = "$ $\leq$ " the relation from $\mathbb{Z}$ to $\mathbb{Z}$.

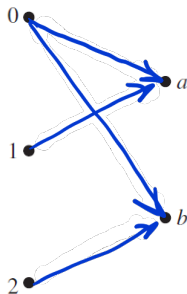
Then $1 \leq 4$ and $5 \not\leq -3$. ($>$)

Let $A = \{0, 1, 2\}$, $B = \{a, b\}$ and $R = \{(0, a), (0, b), (1, a), (2, b)\}$

Then $0 R a$ and $0 R b$ for instance, but $1 \not R b$, since $(1, b) \notin R$

Displaying relations

If $A = \{0, 1, 2\}$, $B = \{a, b\}$ and $R = \{(0, a), (0, b), (1, a), (2, b)\}$ then R can be displayed using arrows or in table:



*This
ex of
relation
which
is not
a function*

R	a	b
0	✓	✓
1	✓	
2		✓

Any function gives a relation (graph of this function), but not any relation gives a function.

Relations on a set

A **relation on a set A** is a relation from A to A .

Let $A = \{1, 2, 3, 4\}$ and $R = \{(a, b) \mid a \text{ divides } b\}$ be the relation on A .
Then $R = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}$.

$(3, 4) \notin R$

Consider relations on the set of integers:

$$R_1 = \{(a, b) \mid a \leq b\}$$

$$R_2 = \{(a, b) \mid a > b\}$$

$$R_3 = \{(a, b) \mid a = b \text{ or } a = -b\}$$

$$R_4 = \{(a, b) \mid a = b\}$$

$$R_5 = \{(a, b) \mid a = b + 1\}$$

$$R_6 = \{(a, b) \mid a + b \leq 3\}$$

Then $(1, 1)$ belongs to R_1, R_3, R_4, R_6

$(1, 2)$ belongs to R_1, R_6

$(2, 1)$ belongs to R_2, R_5, R_6

$(1, -1)$ belongs to R_2, R_3, R_6

$$|\tilde{A} \times \tilde{A}| = n^2$$
$$|\mathcal{P}(A^2)| = 2^{n^2}$$

The number of all possible subsets of A^2

If A is finite with n elements, then the number of all possible relations on A is 2^{n^2}

Properties of relations

A relation R on a set A is called **reflexive** if $(a, a) \in R$ for every $a \in A$.

Which of the following relations on integers is reflexive?

$$R_1 = \{(a, b) \mid a \leq b\} \quad (a, a) \in R_1, \text{ since } a \leq a \Rightarrow \text{yes}$$

$$R_2 = \{(a, b) \mid a > b\} \quad (a, a) \notin R_2, \text{ since } a > a \Rightarrow \text{no}$$

$$R_3 = \{(a, b) \mid a = b \text{ or } a = -b\} \quad (a, a) \in R_3, \text{ since } a = a, \text{ yes}$$

$$R_4 = \{(a, b) \mid a = b\} \quad (a, a) \in R_4, \text{ yes}$$

$$R_5 = \{(a, b) \mid a = b + 1\} \quad (a, a) \notin R_5, \text{ since } a \neq a + 1, \text{ no}$$

$$R_6 = \{(a, b) \mid a + b \leq 3\} \quad (a, a) \in R_6, \text{ for } 2a \leq 3, a \leq 1.5$$

$$R_7 = \{(a, b) \mid a \text{ divides } b\} \quad \text{if } a = 2, \text{ then } (a, a) \notin R_6, \text{ no}$$

Answer: R_1, R_3, R_4, R_7 $\downarrow$ $(a, a) \in R_7, \text{ since } a \mid a, \text{ yes}$

R_5 [if $a=b+1$ and $b=a+1$, then $a=b$] $\Rightarrow R_5$ antisym.
 R_5 is not sym⁰, since $(2,1) \in R_5$ but $(1,2)$ is not.

A relation R on a set A is called **symmetric** if $(b,a) \in R$ whenever $(a,b) \in R$, for all $a,b \in A$.

A relation R on a set A such that for all $a,b \in A$ [if $(a,b) \in R$ and $(b,a) \in R$, then $a=b$] is called **antisymmetric**.

Which of the following relations on integers is symmetric or antisymmetric?

$R_1 = \{(a,b) \mid a \leq b\}$ is if $a \leq b$ and $b \leq a$ then $a=b$,
 $R_2 = \{(a,b) \mid a > b\}$ is it is not sym, antisym.

$R_3 = \{(a,b) \mid a = b \text{ or } a = -b\}$ is since $(1,2) \in R$, but

$R_4 = \{(a,b) \mid a = b\}$ is R_2 is not sym, $(2,1) \notin R$,

$R_5 = \{(a,b) \mid a = b+1\}$ is $(2,1) \in R_2$ but $(1,2) \notin R_2$

$R_6 = \{(a,b) \mid a+b \leq 3\}$ is but antisym.

$R_7 = \{(a,b) \mid a \text{ divides } b\}$ is { if $a > b$ and $b > a$, then $a=b$ } $\Rightarrow$

R_3, R_4 are sym.

R_5 is not antisym

since if $a=-b$, then $(a,b) \in R_3, (b,a) \in R_3$ but $a \neq b$

R_4 is antisym.

A relation R on a set A is called **transitive** if $(a, b) \in R$ and $(b, c) \in R$ then $(a, c) \in R$ for all $a, b, c \in A$.

Which of the following relations on integers is transitive?

- $R_1 = \{(a, b) \mid a \leq b\}$ is if $a \leq b$ and $b \leq c$, then $a \leq c$, yes
 $R_2 = \{(a, b) \mid a > b\}$ is if $a > b$ and $b > c$, then $a > c$, yes
 $R_3 = \{(a, b) \mid a = b \text{ or } a = -b\}$ is If $a R_3 b$ and $b R_3 c \Rightarrow a R_3 c$
 $R_4 = \{(a, b) \mid a = b\}$ is yes $a = b \text{ or } a = -b$ $b = c \text{ or } b = -c$ yes
 $R_5 = \{(a, b) \mid a = b + 1\}$ is if $a = b + 1$ and $b = c + 1$, then $a = c + 2$
 $R_6 = \{(a, b) \mid a + b \leq 3\}$ is if $a + b \leq 3$ and $b + c \leq 3$, then $a + c \leq 3$?
 $R_7 = \{(a, b) \mid a \text{ divides } b\}$ is then it is not nec. $a + c \leq 3$ no
 $a = 2, b = 1, c = 2$ then $a + c = 4 \not\leq 3$, no

Answer Q1

$\swarrow$ if $a \mid b$ and $b \mid c \Rightarrow b = k_1 a$ and $c = k_2 b$
 $c = k_2 b = k_2 k_1 a \Rightarrow a \mid c \Rightarrow a R_7 c$
 $\Rightarrow$ yes.

Homework:

- Solve all odd exercises at p. 581 until #43.

Representing relations (matrices)

Let $A = \{1, 2, 3, 4\}$ and $B = \{1, 2, 3\}$, let also
 $R = \{(1, 2), (1, 3), (2, 2), (3, 1), (3, 2), (4, 2), (4, 3)\}$, then

$$M_R = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \end{matrix}$$

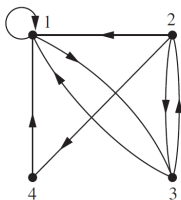
From the matrix representation we can see if the relation is reflexive, symmetric or antisymmetric **$A = \{1, 2, 3\}$ R is on A**

$M_R =$

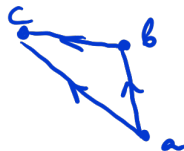
$\begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$ not refl. not sym. not antisym.	$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$ refl. not sym. antisym.	$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ not refl. sym. not ant.	$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$ not refl. not sym. antisym.	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ refl. sym. antisym.
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Representing relations (digraphs)

The directed graph of $R = \{(1, 1), (1, 3), (2, 1), (2, 3), (2, 4), (3, 1), (3, 2), (4, 1)\}$ is

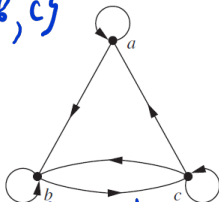


on $\{1, 2, 3, 4\}$

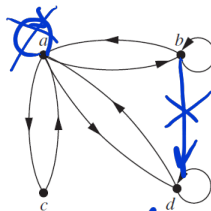


From the digraphs we see if it is reflexive, symmetric, antisymmetric or transitive.

on $\{a, b, c\}$



refl, not sym.
not antisym. trans.



not refl, sym, not antisym.
not trans.

Closure of relations

Reflexive closure of R is the smallest reflexive relation which contains R

Reflexive closure of $R = \{(1, 1), (2, 3), (2, 4), (3, 1)\}$ is ^{on $\{1, 2, 3\}$}
 $\{(1, 1), \underline{(2, 2)}, (2, 3), (2, 4), (3, 1), \underline{(3, 3)}, \underline{(4, 4)}\}$

Symmetric closure of R is the smallest symmetric relation which contains R

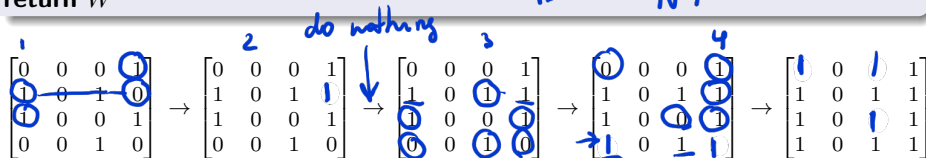
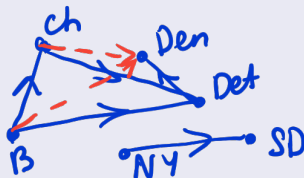
Symmetric closure of $R = \{(1, 1), (2, 3), (2, 4), (3, 1)\}$ is
 $\{(1, 1), \underline{(1, 3)}, (2, 3), (2, 4), (3, 1), \underline{(3, 2)}, \underline{(4, 2)}\}$

Transitive closure of R is the smallest transitive relation which contains R

Transitive closure of $R = \{(1, \underline{1}), \underline{(2, 3)}, (2, 4), (3, \underline{1})\}$ is ^{if (a, b) and (b, c) then (a, c)}
 $\{(1, 1), \underline{(2, 1)}, (2, 3), (2, 4), (3, 1)\}$

Warshall's algorithm

procedure Warshall ($\underline{M_R} : n \times n$ zero-one matrix) *on A, |A|=n*
 $W := M_R$
for $k := 1$ **to** n
 for $i := 1$ **to** n
 for $j := 1$ **to** n
 $w_{ij} := w_{ij} \vee (w_{ik} \wedge w_{kj})$
return W



Example

A computer network has data centers in Boston, Chicago, Denver, Detroit, New York, and San Diego. There are direct, one-way telephone lines from Boston to Chicago, from Boston to Detroit, from Chicago to Detroit, from Detroit to Denver, and from New York to San Diego. Let R be the relation containing (a, b) if there is a telephone line from the data center in a to that in b . How can we determine if there is some (possibly indirect) link composed of one or more telephone lines from one center to another?

Homework:

- Solve exercises # 1,3,19,21,23,25,27 at p. 596-597.
- Solve exercises # 1,3,5,7,9,11,25,27,29 at p. 606-607.

Answer Q2

Equivalence relations

(Paragraph 9.5)

A relation on a set A is called an **equivalence relation** if it is reflexive, symmetric, and transitive.

Two elements a and b that are related by an equivalence relation are called **equivalent**.

The notation $a \sim b$ is used to denote that a and b are equivalent elements with respect to a particular equivalence relation.

If $R = \{(a, b) \mid a = b\}$ is equivalence relation and $a \sim b$ if and only if $a = b$.

If $R = \{(a, b) \mid a = b \text{ or } a = -b\}$ is equivalence relation and $a \sim b$ if and only if $a = b$ or $a = -b$.

Let aRb be related if $a - b$ is an integer. Then R is equivalence relation (proof).

Let $R = \{(a, b) \mid a \equiv b \pmod{m}\}$, then R is equivalence relation (proof).

Two identifiers (names of variables in some programming language) are related by relation R_8 if when they agree with the first 8 characters. Then R_8 is equivalence relation (proof).

R_3 is equivalence relation on the set of bit strings such that aR_3b if a and b start with the same three bits (proof).

Why $R = \{(a, b) \mid a \text{ divides } b\}$ is not equivalence relation?

Answer Q3

Equivalence Classes and Partitions

Let R be an equivalence relation on a set A . The set of all elements that are related to an element a of A is called the **equivalence class** of a .

The equivalence class of a with respect to R is denoted by $[a]_R$. So

$$[a]_R = \{s \mid (a, s) \in R\}.$$

If $R = \{(a, b) \mid a = b\}$, then $[a]_R = \{a\}$.

Let aRb be related if $a - b$ is an integer. Then

$$[5.2]_R = \{s \mid 5.2 - s \in \mathbb{Z}\} = \{0.2, 1.2, 2.2, -0.8, -1.8, -2.8, \dots\}$$

Let $R = \{(a, b) \mid a \equiv b \pmod{7}\}$, then

$$[5]_R = \{-2, 5, 12, 19, \dots\}$$

$$[\text{number_of_something}]_{R_8} = [\text{number_of_student}]$$

$$[110]_{R_3} = \{110, 1101, 1100, 1101101, \dots\}$$

Answer Q4

Theorem

Let R be an equivalence relation on a set A . These statements for elements a and b of A are equivalent:

(i) aRb

(ii) $[a] = [b]$

(iii) $[a] \cap [b] \neq \emptyset$

(i) $\xrightarrow{\checkmark}$ (ii) $\xrightarrow{\checkmark}$ (iii) $\rightarrow$ (i) way of proof.

proof: (i) $\rightarrow$ (ii) Assume $aRb, c \in [a]$, then $aRc \Rightarrow cRa$

cRa and $aRb \xRightarrow{+r} cRb \Rightarrow c \in [b] \Rightarrow [a] \subset [b]$

similarly, $[b] \subset [a] \Rightarrow [a] = [b]$

(ii) $\rightarrow$ (iii) Assume $[a] = [b] \Rightarrow [a] \cap [b] \neq \emptyset$

(iii) $\rightarrow$ (i) $[a] \cap [b] \neq \emptyset \Rightarrow \exists c$ s.t. $c \in [a]$ and $c \in [b]$
 $\Rightarrow cRa$ and $cRb \xRightarrow{+r, \text{sym.}} aRc$ and $cRb \Rightarrow aRb$ $\square$

$A_1/A_2/A_3/A_4/A_5 \dots A$ $A_i \cap A_j = \emptyset$ and $\bigcup_{j=1}^5 A_j = A \leftarrow \text{partition of } A$

We have that $\bigcup_{a \in A} [a]_R = A$ and if $a \not\sim b$ then $[a] \cap [b] = \emptyset$.

Partition of a set

We have that $\bigcup_{a \in A} [a]_R = A$ and if $a \not\sim b$ then $[a] \cap [b] = \emptyset$.

Theorem

Let R be an equivalence relation on a set A .

Then the equivalence classes of R form a **partition** of A .

What is the partition of $\mathbb{Z} = \{ \dots, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, \dots \}$ on the set of integers. $\mathbb{Z} = \{0, 4, 8, 12, \dots\} \cup [1] \cup [2] \cup [3]$
 R_3 on the set of bit strings. *try to understand partition!*

Theorem

Conversely, given a partition $\{A_i \mid i \in I\}$ of the set A , there is an equivalence relation R that has the sets A_i , $i \in I$, as its equivalence classes.

List the ordered pairs in the equivalence relation R produced by the partition $A_1 = \{1, 2, 3\}$, $A_2 = \{4, 5\}$, and $A_3 = \{6\}$ of $A = \{1, 2, 3, 4, 5, 6\}$. $(4, 5), (5, 4)$
Answer Q5 $R = \{(i, i) \mid i \in A\} \cup \{(1, 2), (2, 1), (1, 3), (3, 1), (2, 3), (3, 2)\}$

Homework:

- Solve exercises # 1,3,5,7,9,13,15,19,27,29,37,39,41,43,45,47,57 at p. 615-617.

Partial ordering

(Paragraph 9.6)

A relation R on a set S is called a **partial order** if it is reflexive, antisymmetric, and transitive.

A set S together with a partial order R is called a **partially ordered set**, or poset, and is denoted by (S, R) .

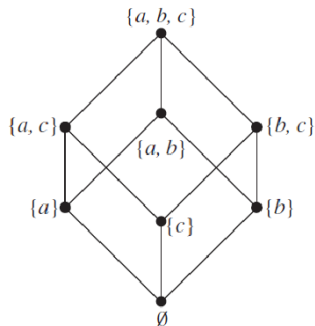
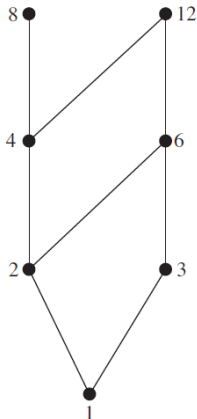
The "greater than or equal" relation ($\geq$) is a partial ordering on the set of integers. So $(\mathbb{Z}, \geq)$ is a poset. *total order*
 $(\mathbb{Z}^+, |)$ is a poset. *partial order*
 $(\mathcal{P}(S), \subseteq)$ is a poset. *$\{1, 2\}$ and $\{2, 3\}$ not compar. partial order.*

The elements $a, b \in S$ are called **comparable** if either aRb or bRa . When a and b are such that neither aRb nor bRa , a and b are **incomparable**.

If for all $a, b \in S$, a and b are comparable, then R is called total order.

In the poset $(\mathbb{Z}^+, |)$, are the integers 3 and 9 comparable? *3|9, yes*
Are 5 and 7 comparable? *no, since 5 $\nmid$ 7 and 7 $\nmid$ 5.*
Which of the relations above is a total order? *○*

Hasse Diagrams



$$(\{1, 2, 3, 4\}, \leq), \quad (\{1, 2, 3, 4, 6, 8, 12\}, |), \quad (\mathcal{P}(\{a, b, c\}), \subseteq)$$

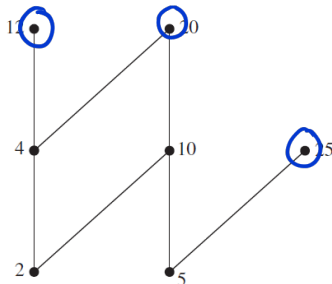
Maximal and Minimal Elements

Let $(S, \preceq)$ be a poset.

The element $a \in S$ is **maximal** if for any $b \in S$ comparable with a , we have $b \preceq a$.

The element $a \in S$ is **minimal** if for any $b \in S$ comparable with a , we have $a \preceq b$.

Which elements of the poset $(\{2, 4, 5, 10, 12, 20, 25\}, |)$ are maximal, and which are minimal?



From the Hasse diagram
the maximal elements are
the minimal elements are

12, 20, 25
2, 5

we see that

Topological sorting

procedure topological sort $((S, \preceq)$: finite poset)

$k := 1$

while $S \neq \emptyset$

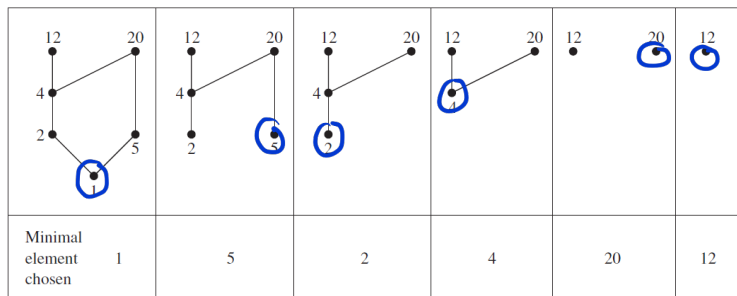
$a_k := a$ minimal element of S

$S := S \setminus \{a_k\}$

$k := k + 1$

return $a_1, a_2, \dots, a_n$

A topological sort of $(\{1, 2, 4, 5, 12, 20\}, |)$ is



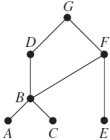
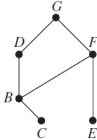
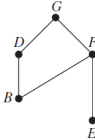
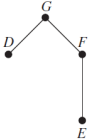
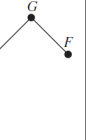
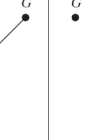
Scheduling of projects

A development project requires the completion of seven tasks A, B, C, D, E, F, G . Some of these tasks can be started only after other tasks are finished.

Set $X \preceq Y$ if task Y cannot be started until task X has been completed.

This relation is given as follows $A, C \preceq B$, $B, E \preceq F$, $B \preceq D$, and $D, F \preceq G$. Find an order in which these tasks can be carried out to complete the project.

A topological sort is

						
Minimal element chosen A	C	B	E	F	D	G

A, C, B, E, F, D, G is one possible order.

Answer Q6

Homework:

- Solve #1,3,5,7,9,11,15,21,23,25,27,33 a) and b), 35 a) and b), 67 at p. 630-633.